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Industrial Internship Report on

" Forecasting of Smart city traffic patterns "

Prepared by

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Executive Summary

This report provides details of the Industrial Internship provided by Upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt. Ltd. (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to complete the project along with the report within six weeks.

My project was "Forecasting of Smart City Traffic Patterns Using Machine Learning." The objective of this project was to develop a machine learning model capable of forecasting traffic patterns at different city junctions using historical traffic data. The project involved data preprocessing, feature engineering, exploratory data analysis (EDA), model development using the Random Forest Regressor, and prediction of traffic volume. The model was trained using the training dataset and tested on the testing dataset to forecast future traffic patterns.

This internship gave me a very good opportunity to gain exposure to real industrial problems and implement practical machine learning solutions. It was an enriching learning experience that enhanced both my technical and analytical skills.

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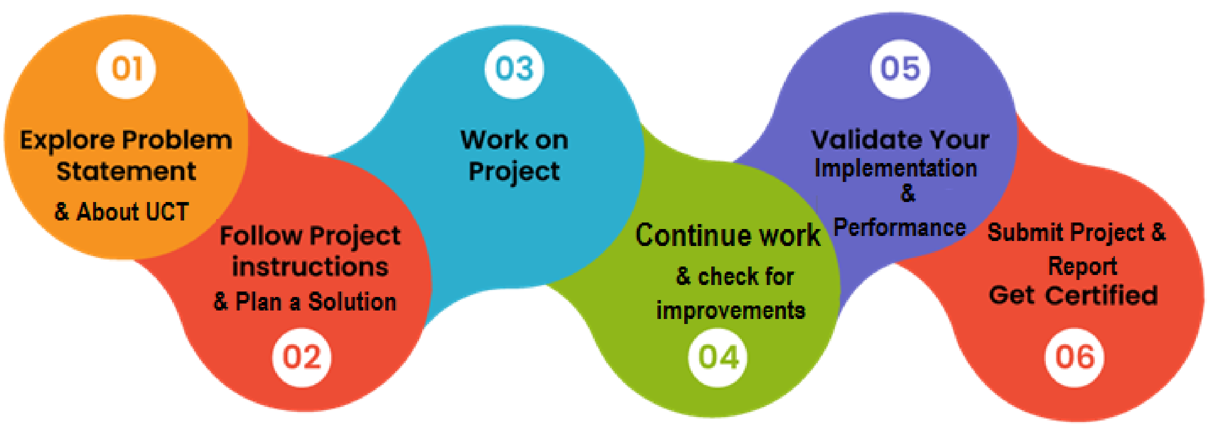
1 Preface

The six-week Industrial Internship organized by **Upskill Campus** and **The IoT Academy** in collaboration with **UniConverge Technologies Pvt. Ltd. (UCT)** provided me with valuable practical exposure to Data Science and Machine Learning. The internship helped me understand how machine learning techniques can be applied to solve real-world industrial problems while improving my technical and analytical skills.

As part of the internship, I worked on the project **"Forecasting of Smart City Traffic Patterns Using Machine Learning."** The project focused on predicting traffic patterns at different city junctions using historical traffic data. The implementation involved data preprocessing, feature engineering, exploratory data analysis (EDA), model training using the Random Forest algorithm, and prediction of future traffic patterns.



The internship was a valuable learning experience that improved my understanding of data preprocessing, feature engineering, machine learning model development, and performance evaluation. I sincerely thank **Upskill Campus**, **The IoT Academy**, **UniConverge Technologies Pvt. Ltd. (UCT)**, my mentors, and faculty members for their continuous guidance and support. I encourage my juniors and peers to participate in such internship programs as they provide valuable practical exposure and industry-oriented learning.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

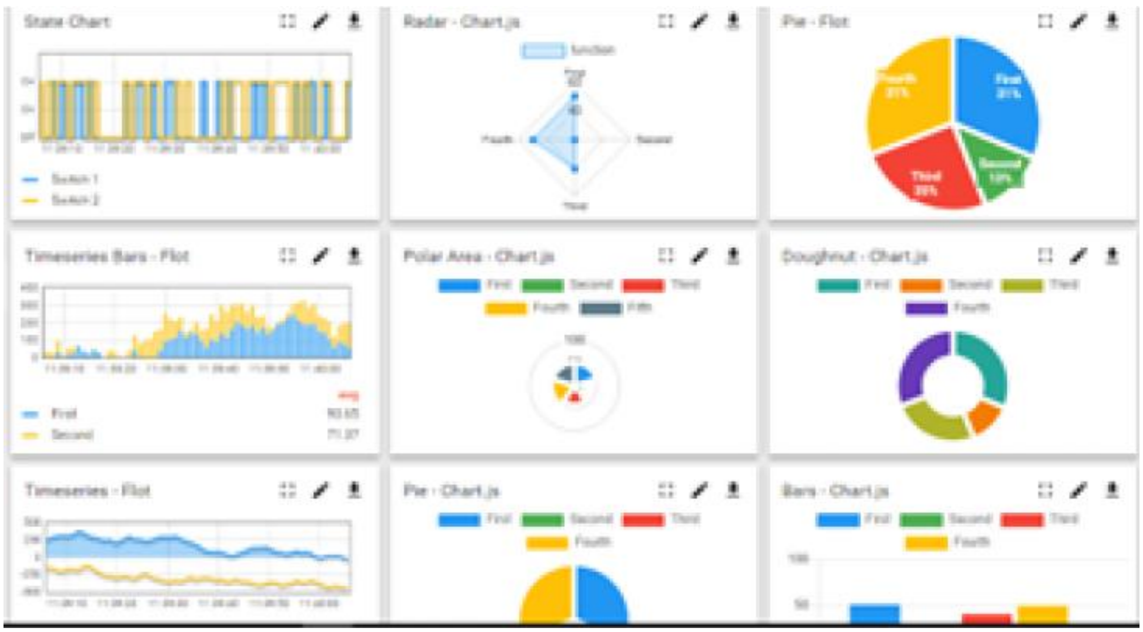
i. UCT IoT Platform ( )

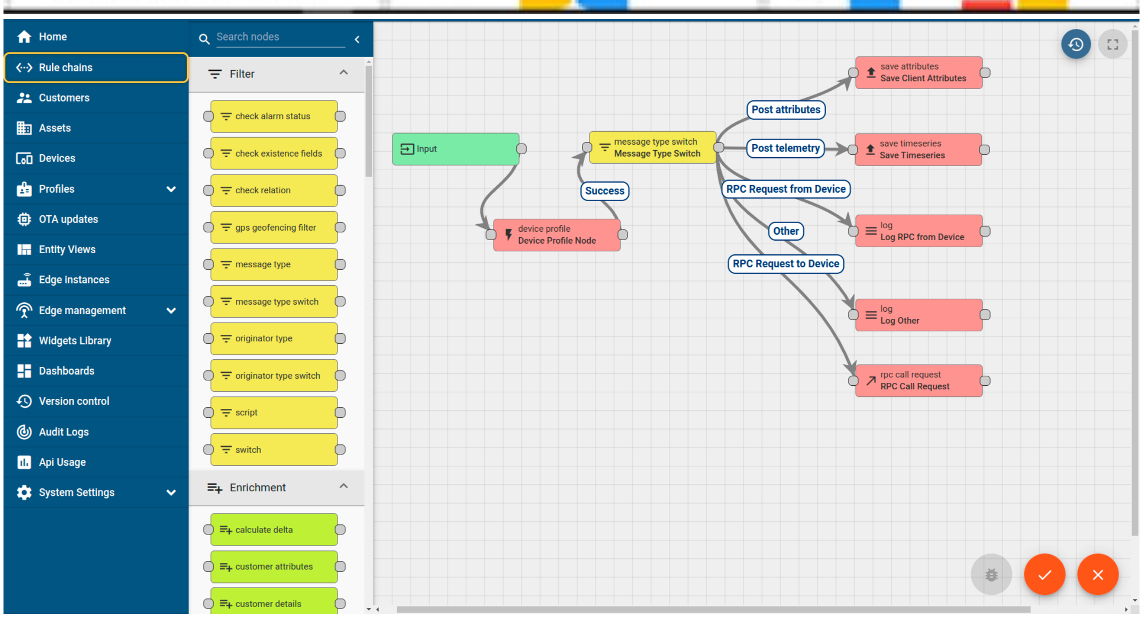
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine




Home
Rule chains
Customers
Assets
Devices
Profiles
OTA updates
Entity Views
Edge instances
Edge management
Widgets Library
Dashboards
Version control
Audit Logs
Api Usage
System Settings
Search nodes
Filter
check alarm status
check existence fields
check relation
gps geofencing filter
message type
message type switch
originator type
originator type switch
script
switch
Enrichment
calculate delta
customer attributes
customer details
Input
device profile
Device Profile Node
message type switch
Message Type Switch
Success
Post attributes
Post telemetry
RPC Request from Device
Other
RPC Request to Device
save attributes
Save Client Attributes
save timeseries
Save Timeseries
log
Log RPC from Device
log
Log Other
rpc call request
RPC Call Request

ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

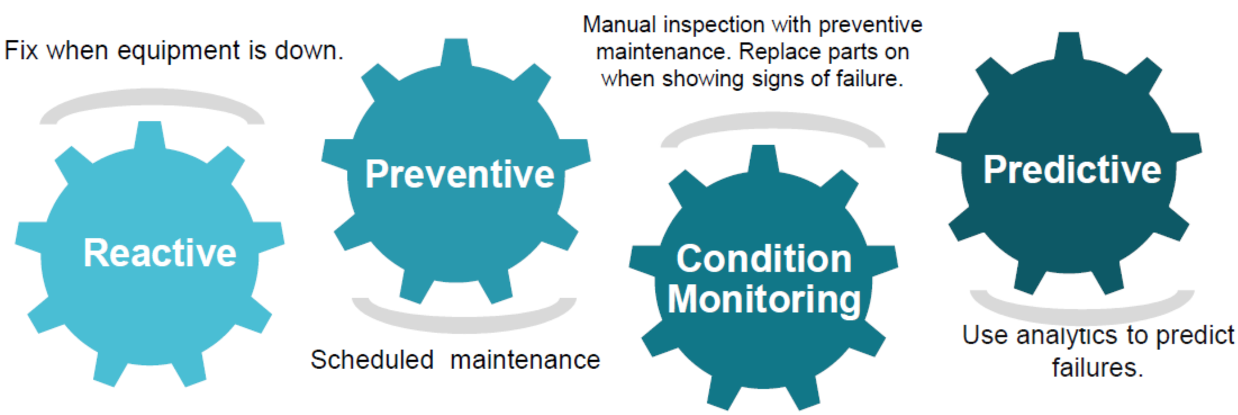
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

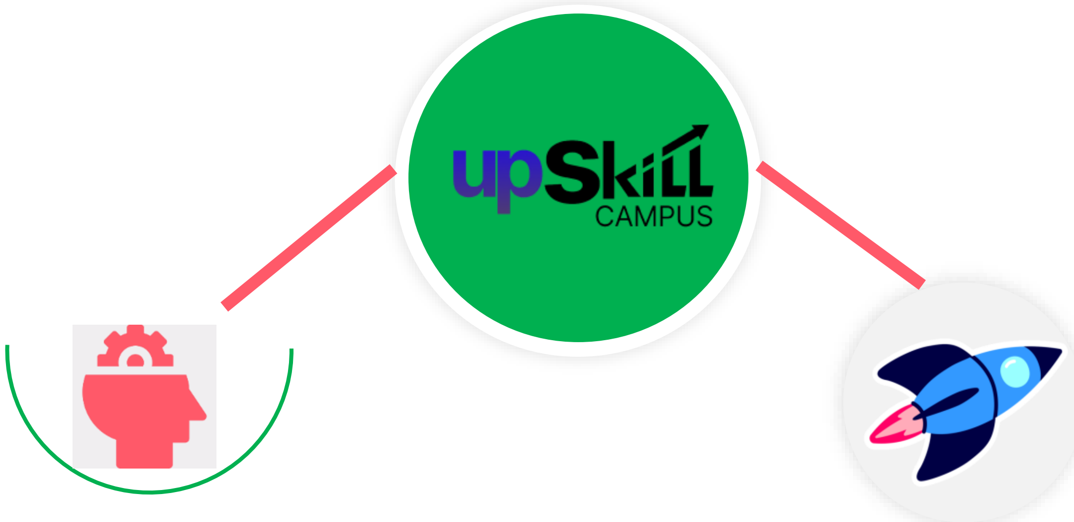
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/

Career growth/upskilling
- Interview Preparation and skill building
- upskilling Courses
- Skill Assessment
- Profile building

Professional networking
- Alumni Connections
- Mentorship
- Discussion/QA forum

Collaboration platform
- Project collaboration
- Discussion forum
- Tech updates

Job/internship platform
- Job portal
- Internship portal
- Freelancing projects

The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.3 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

2.4 Reference

[1] Upskill Campus Learning Materials

[2] The IoT Academy Learning Resources

[3] Dataset provided by UniConverge Technologies Pvt. Ltd. (UCT)

2.5 Glossary

Terms	Acronym
AI	Artificial Intelligence
ML	Machine Learning
EDA	Exploratory Data Analysis
MAE	Mean absolute error
Linear Regression	A supervised machine learning algorithm used to predict the continuous values

3 Problem Statement

Rapid urbanization has significantly increased the number of vehicles on city roads, resulting in traffic congestion and delays. Efficient traffic management has become an essential requirement for developing smart cities. Predicting traffic patterns helps government authorities understand traffic flow, identify peak traffic periods, and improve infrastructure planning and traffic management.

The assigned problem statement focuses on **"Forecasting of Smart City Traffic Patterns Using Machine Learning."** The objective is to develop a machine learning model capable of forecasting traffic volume at different city junctions using historical traffic data. Accurate traffic forecasting supports better planning, efficient traffic control, and improved transportation services.

In this project, historical traffic data was preprocessed and analyzed using Python. Feature engineering techniques were applied to extract useful information from the datetime column, followed by Exploratory Data Analysis (EDA). A **Random Forest Regressor** model was trained using the training dataset and used to predict traffic patterns for the testing dataset.

4 Existing and Proposed solution

4.1 Existing Solution

Traditional traffic management systems mainly rely on manual monitoring, fixed traffic signals, and historical observations. These methods are often unable to accurately predict future traffic conditions and cannot efficiently analyze large volumes of traffic data. As cities continue to grow, conventional approaches become less effective in managing traffic congestion and infrastructure planning.

- **Limitations**
- Manual monitoring is time-consuming.
- Difficult to analyze large traffic datasets.
- Limited prediction capability.
- Cannot effectively identify complex traffic patterns.
- Less suitable for smart city applications.

4.2 Proposed Solution

The proposed solution is a **Machine Learning-based Traffic Forecasting System** developed using the **Random Forest Regressor** algorithm. The system uses historical traffic data collected from different city junctions to forecast future traffic volume.

The implementation includes data preprocessing, datetime conversion, feature engineering, exploratory data analysis, model training, prediction, and evaluation. The Random Forest model effectively captures patterns in historical traffic data and provides reliable traffic forecasts that can assist in urban traffic planning.

- **Value Addition**
- Automated traffic forecasting.

- Better understanding of traffic trends.
- Supports infrastructure planning.
- Assists in reducing traffic congestion.
- Provides data-driven insights for smart city development.

4.1 Code submission : https://github.com/joshikacrisane05-tech/upskillcampus/blob/main/ForecastingTrafficPatterns.ipynb

4.2 Report submission : https://github.com/joshikacrisane05-tech/upskillcampus/blob/main/ForcastingTrafficPatterns_Joshika_USC_UCT.pdf

5 Proposed Design/ Model

The proposed system predicts traffic patterns using historical traffic data collected from multiple city junctions. The implementation follows a structured workflow beginning with data collection and preprocessing, followed by feature engineering, model training, prediction, and evaluation.

Workflow

- Training Dataset
- Testing Dataset
- Data Preprocessing
- Datetime Conversion
- Feature Engineering
- Exploratory Data Analysis (EDA)
- Random Forest Regressor
- Prediction
- Performance Evaluation

5.1 High Level Diagram (if applicable)

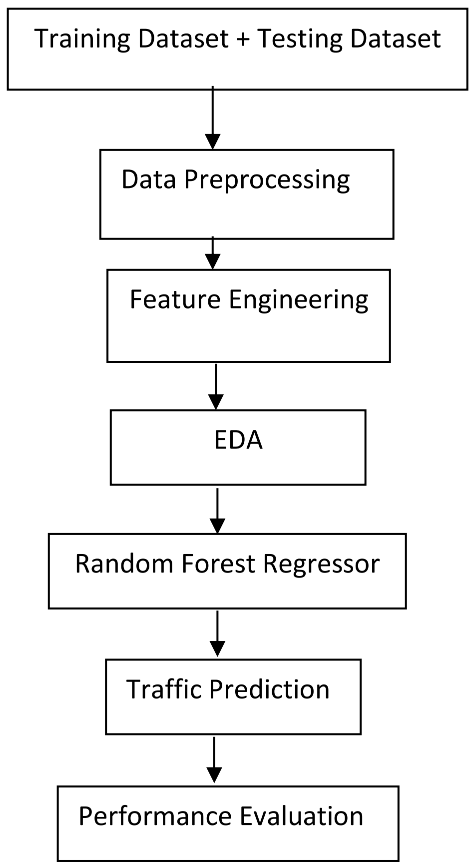


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

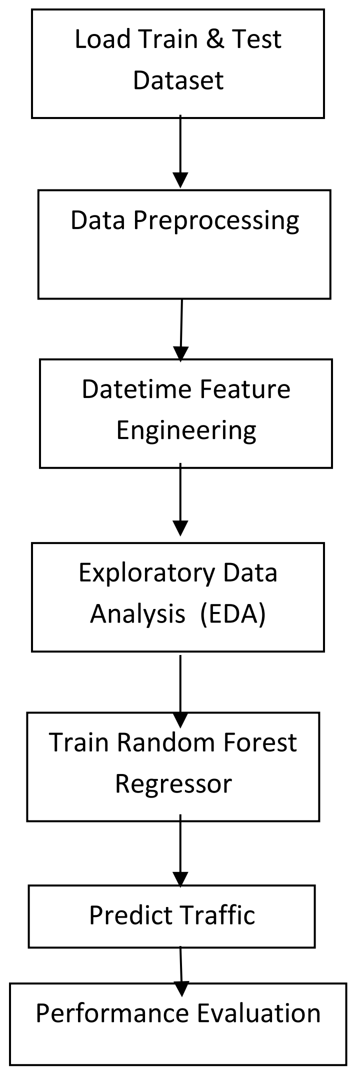


Figure 2: LOW LEVEL DIAGRAM OF THE SYSTEM

6 Performance Test

The developed Random Forest model was evaluated using historical traffic data collected from different city junctions. The training dataset was used to train the model, while the testing dataset was used to forecast future traffic patterns. Datetime feature extraction improved the model by providing additional temporal information such as year, month, day, hour, and day of the week.

The model successfully generated traffic forecasts for the testing dataset. Proper preprocessing and feature engineering improved the model's ability to learn traffic trends and generate reliable predictions.

6.1 Test Plan / Test Cases

Test Case	Expected Result	Status
Load training dataset	Successfully loaded	Passed
Load testing dataset	Successfully loaded	Passed
Datetime conversion	Converted successfully	Passed
Feature engineering	New features created	Passed
Train Random Forest model	Model trained successfully	Passed
Predict traffic volume	Predictions generated	Passed
Display prediction graph	Graph generated successfully	Passed

6.2 Test Procedure

1. Load the training and testing datasets.
2. Convert the datetime column into datetime format.
3. Extract Year, Month, Day, Hour, and Day of Week.
4. Perform Exploratory Data Analysis (EDA).
5. Prepare input features and target variable.
6. Train the Random Forest Regressor using the training dataset.
7. Predict traffic volume for the testing dataset.
8. Visualize the predicted traffic patterns.

6.3 Performance Outcome

The Random Forest Regressor successfully learned traffic patterns from the historical training data and generated traffic predictions for the testing dataset. Feature engineering improved the model's understanding of temporal patterns, enabling better forecasting of traffic volume. The project achieved its objective of developing a machine learning-based traffic forecasting system for smart city applications

7 My learnings

The Industrial Internship provided me with valuable practical experience in Machine Learning and Data Science. During this project, I learned how to preprocess real-world traffic datasets, perform feature engineering, analyze data through exploratory data analysis (EDA), and develop a traffic forecasting model using the Random Forest algorithm.

I also improved my understanding of Python programming and machine learning libraries such as Pandas, NumPy, Matplotlib, and Scikit-learn. This project enhanced my analytical thinking, problem-solving skills, and confidence in applying machine learning techniques to solve real-world problems. The knowledge and practical experience gained through this internship will be highly beneficial for my future career in Data Science and Artificial Intelligence.

8 Future work scope

The current system successfully forecasts traffic patterns using historical traffic data. In the future, the model can be enhanced by incorporating additional factors such as weather conditions, public holidays, festivals, road construction, accidents, and special events that influence traffic flow.

Advanced forecasting techniques such as **XGBoost**, **LSTM (Long Short-Term Memory)**, or other deep learning models can be explored and compared with the Random Forest algorithm to improve prediction accuracy. The system can also be integrated into a real-time smart city traffic management platform to provide live traffic predictions and support intelligent transportation planning.